Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

MICROPROCESSORS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated.
(2) Due credit will be given to neatness and adequate dimensions.
(3) Use of electronic gadgets leaving scientific calculator is not allowed.
(4) Assume suitable data wherever necessary.

1. (a) Recall and explain any three of the following instructions in detail in regards to addressing mode :

   (a) LXI Sp, 3FFFH

   (b) XCHG

   (c) CC 8000H

   (d) PUSH PSW 5(CO1,2)

   (b) Using your knowledge of programming write an assembly language program to find a specific byte of data from a group of 200 bytes stored at location 3000H onwards. The count of repetition should be stored at 5000H. 5(CO2)

2. (a) Illustrate and explain the timing diagram of JMP 8000h. 5(CO1,4)

   (b) Demonstrate stack operation in 8085 microprocessor with example program. 5(CO1,2,4)

3. (a) Integrate a Seven Segment Display in Common Cathode with 8255 at port A and 8085 where chip select is at 40h. Draw interfacing diagram for the same and write a program to display 0 to 9 continuously with a 50 ms delay. Show all calculations considering a crystal of 3 MHz. 10(CO2,3)

4. (a) Clarify requirement of BIU and EU in regards to pipelining with neat diagram. 5(CO4)

   (b) Compare important features of Intel 286, 386 and 486 microprocessor. 5(CO4)

5. (a) Explain addressing mode of 8086 Microprocessor with examples. 5(CO3)

   (b) Write an 8086 assembly language program to Interchange two blocks of data. 5(CO1,4)

6. (a) Deconstruct 8086 Assembly language Language program to find smallest number from a group of 100 bytes of data. 5(CO2)

   (b) Demonstrate use of INT 10h and INT 16h to display string "HELLO WORLD" using 8086 microprocessor. 5(CO2)